ShopSphere

Product Requirements Document — Multi-Vendor E-commerce Marketplace (v4.1)

Status: Approved for Development | Owner: Marketplace Platform Team | Version 4.1 | Classification: Internal

1. Executive Summary

ShopSphere is a multi-vendor e-commerce marketplace connecting independent sellers with shoppers through a unified storefront, checkout, and fulfillment experience. Sellers list and manage inventory, shoppers browse, search, and purchase across thousands of vendors with a single cart and checkout, and ShopSphere handles payment collection, tax calculation, fraud screening, and payout splitting to sellers. This document defines requirements for the v4.1 release, which introduces a personalized recommendation engine and same-day local delivery partnerships.

2. Problem Statement & Objectives

Independent sellers struggle to reach shoppers without building their own storefront, payment, and logistics infrastructure. ShopSphere gives sellers instant access to a large shopper base while shoppers get a single trusted checkout across many vendors.

- Grow active seller base by 40% while keeping seller onboarding-to-first-sale under 3 days.
- Achieve sub-2-second search results for catalog queries at peak traffic (10x average load).
- Reduce checkout cart abandonment by improving payment and tax-calculation latency.
- Launch personalized product recommendations driving a measurable lift in average order value.

3. Target Users & Personas

Persona	Description	Primary Needs
Shopper	End consumer browsing and buying	Fast search, trusted checkout, order tracking
Seller	Independent vendor listing products	Inventory tools, sales analytics, timely payouts
Marketplace Ops	Internal trust & safety / catalog team	Listing moderation, dispute resolution tools
Finance / Payouts Admin	Internal finance team	Payout reconciliation, tax reporting, refund ove

4. Scope

4.1 In Scope

- Multi-vendor product catalog with search, filtering, and personalized recommendations.

- Unified shopping cart and checkout spanning multiple sellers in a single order.
- Real-time tax calculation and multi-carrier shipping rate quotes at checkout.
- Seller inventory management, order fulfillment status, and payout dashboards.
- Fraud screening on orders prior to payment capture.
- Post-purchase order tracking, returns, and refund workflows.

4.2 Out of Scope (this release)

- In-house last-mile delivery fleet (same-day delivery is via third-party partners only).
- Live-stream shopping / social commerce features.
- Business-to-business bulk ordering and net-terms invoicing.

5. Functional Requirements

5.1 Catalog & Search

- FR-1: System shall index seller-submitted product listings and make them searchable within 5 minutes of publish.
- FR-2: System shall support faceted search (category, price, rating, shipping speed) with sub-2-second P95 response time.
- FR-3: System shall surface personalized product recommendations on the homepage and product detail pages.

5.2 Cart & Checkout

- FR-4: System shall support a single cart containing items from multiple sellers, split into separate sub-orders at fulfillment.
- FR-5: System shall calculate applicable sales tax per shipping jurisdiction at checkout.
- FR-6: System shall present real-time shipping rate and delivery-window options from connected carriers.
- FR-7: System shall screen each order for fraud signals before authorizing payment capture.

5.3 Seller Management

- FR-8: System shall allow sellers to manage inventory levels, pricing, and listing content via a self-service portal.
- FR-9: System shall calculate and schedule seller payouts net of commission, weekly by default.
- FR-10: System shall notify sellers of low-stock and order-fulfillment SLA breaches.

5.4 Orders, Returns & Support

- FR-11: System shall provide shoppers real-time order and shipment tracking per sub-order.
- FR-12: System shall support return initiation, refund processing, and seller-side dispute workflows.

6. System Architecture Overview

ShopSphere follows a layered microservices architecture (see attached Architecture Diagram). Shoppers and sellers connect through dedicated web/mobile clients and a seller portal, all terminating at a CDN/WAF and API gateway. The Cart & Checkout Service coordinates with the Product Catalog & Search Service, Recommendation Engine, and Order Management Service, and calls out to the Payment Gateway only at the moment of purchase, keeping raw payment data out of ShopSphere's own data stores. The Seller & Inventory Service manages listings and stock levels independently, publishing change events to a shared event bus consumed by Search indexing and the Notification Service. Order and payment events flow asynchronously to keep seller payout calculations eventually consistent with fulfillment status.

- Client Layer: Shopper Web App (Next.js), Shopper Mobile App (iOS/Android), Seller Portal.
- Edge/Perimeter: CDN/WAF, API Gateway (authn, rate limiting), Load Balancer.
- Application Services: Auth & Account Service, Product Catalog & Search Service, Recommendation Engine, Cart & Checkout Service, Order Management Service, Seller & Inventory Service.
- Data Stores: Catalog Search Index (Elasticsearch), Orders DB (PostgreSQL), Redis Cache (cart/session), Product Image Store (S3), Order Event Bus (Kafka).
- External Dependencies: Payment Gateway (Stripe/Adyen), Tax Calculation Service (Avalara), Shipping/Logistics Carriers, Fraud Screening Provider (Sift).

7. Data Classification & Sensitive Data Inventory

Data Type	Classification	Where Stored	Regulatory Driver
Shopper payment card data	Restricted / PCI	Never stored; tokenized via Payment Gateway	PCI DSS
Shopper account & shipping info	PII	Auth & Account Service, Orders DB (encrypted)	GDPR / CCPA
Seller banking / payout details	Restricted / Financial	Seller & Inventory Service, encrypted column-level	GLBA-equivalent, PCI DSS
Order & transaction history	Internal / Financial	Orders DB	SOX (financial reporting), tax
Product catalog content	Public	Catalog Search Index, Image Store	N/A
Behavioral / browsing data	PII (limited)	Recommendation Engine store	GDPR / CCPA

8. Authentication, Authorization & Session Management

- Shoppers authenticate via email/password or social SSO; guest checkout is supported with post-purchase account linking.
- Sellers require MFA for portal access given exposure to payout and banking configuration.
- Role-based access separates Shopper, Seller, Marketplace Ops (catalog moderation), and Finance Admin (payout/refund authority) permissions.
- Server-to-server integrations (carriers, tax, fraud) use scoped API credentials rotated quarterly.
- Session tokens are short-lived JWTs with refresh rotation; cart state is tied to session, not raw PII.

9. Non-Functional Requirements

Category	Requirement
Availability	99.9% uptime for browse/checkout path; multi-region active-active for catalog reads
Search Latency	P95 search response under 2 seconds at 10x average peak load
Scalability	Support flash-sale traffic spikes of 20x baseline without checkout degradation
Data Consistency	Inventory counts eventually consistent within 5 seconds of a sale
Auditability	All payout, refund, and listing-moderation actions logged for 3+ years

10. Compliance & Regulatory Requirements

- PCI DSS SAQ-A eligibility maintained by never handling raw card data directly.
- GDPR / CCPA data subject rights for shopper accounts and behavioral data.
- State and cross-border sales tax remittance obligations (via Tax Calculation Service).
- Marketplace facilitator liability regulations for tax collection in applicable jurisdictions.

11. Assumptions, Constraints & Dependencies

- Shipping rate accuracy depends on carrier API uptime and seller-provided package dimensions.
- Recommendation quality depends on sufficient browsing/purchase history; cold-start shoppers see popularity-based fallbacks.
- Seller payout timing is constrained by the Payment Gateway's settlement schedule.

12. Open Questions & Risks

- Should same-day delivery partner selection be shopper-visible or automatically optimized?
- Risk: Multi-seller cart splitting adds checkout complexity that could increase abandonment if not well designed.
- Risk: Flash-sale traffic spikes have historically caused search index lag; needs load testing before GA.